FACTORS AFFECTING

Moment of force Torque	a) The magnitude of the force applied (more the force greater the torque) b) The perpendicular distance of the line of action of force from the pivoted point. (more the distance greater the turning effect)
Centre of gravity	Shape i.e. on the distribution of mass on the body
Work done	a) Magnitude of the force applied (more the force more the work) b) Displacement produced by the force.(more the displacement more the work)
Power	a) Amount of work done by the source (more the work more the power) b) Time taken by the source to do the work (more the time less the power)
Efficiency and Mechanical advantage	a) Friction b) Weight of the moving parts of the machine (increase in these factors reduces the efficiency and the MA)
Refractive index	a) Temperature (More the temperature lesser the refractive index) b) Wavelength of light (more the wavelength less the refractive index)
Lateral displacement	a) Thickness of medium (more the thickness more the lateral disp) b) Angle of incidence (increases with the increase in angle of incidence) c) Refractive index (increases with the increase in refractive index)
Angle of deviation(δ)	a) Angle of incidence (as the angle of incidence increases, the angle of deviation first decreases, reaches to the minimum value and then increases) b) Refractive index (more the refractive index more the deviation) c) Colour or wavelength (more the wavelength lesser the deviation)
Shift	a) Refractive index (higher the refractive index, more the shift) b) Colour or wavelength (more the wavelength, lesser the shift)
Critical angle	a) Colour or wavelength of light (more the wavelength of light, more the critical angle) b) Temperature (increase in temperature increases the critical angle)
Power of a lens	a) Focal length (more the focal length lesser the power)
Loudness of a sound	a) Amplitude: Loudness is directly proportional to the square of amplitude. b) Distance from the source: Loudness is inversely proportional to the square of the distance from the source c) Surface area of the vibrating body: More the surface area more the loudness
Pitch	a) Frequency: more the frequency higher the pitch
Resistance of a conductor	a) Length of conductor: More the length of conductor more the resistance b) Area or thickness of a conductor: More the thickness of a conductor lesser the resistance c) Temperature of a conductor: More the temperature, more the resistance
Specific resistance or resistivity	a) Depends on temperature i. For metals, increases with increase in temperature ii. For semiconductors, decreases with increase in temperature iii. For alloys like constantan remains unaffected b) Do not depends on shape and size
EMF of a cell	a) The material of the electrode b) The electrolyte used in cell

FACTORS AFFECTING

Internal resistance of a cell	a) The surface area of electrode: Larger the surface area, lesser the internal resistance b) Distance between the electrodes: More the distance between the electrodes, greater the internal resistance c) Temperature of the electrolyte: Higher the temperature of the electrolyte, less the internal resistance
Heating effect of current	**a)** Directly proportional to the square of current, $H \propto I^2$ **b)** Directly proportional to the resistance of the wire, $H \propto R$ **c)** Directly proportional to the time for which current is passed in the wire $H \propto t$
the quantity of heat absorbed to increase the temperature of the body	a) Mass of the body: More the mass, more is the heat absorbed. $Q \propto m$ b) Rise in temperature: More the rise in temperature, more the heat absorbed. $Q \propto \Delta t$ c) Specific heat capacity (c): $Q \propto c$

NECESSARY CONDITIONS REQUIRED

For body to be in equilibrium	a) The resultant of all the forces acting on a body should be zero b) The algebraic sum of moment of all the forces acting on a body at a point of rotation should be zero.
Work done by a force to be zero	a) When there is **no displacement** b) When the **displacement is normal to the direction of force** ($\theta=90^0$)
No refraction takes place or the light ray pass undeviated	a) When the **angle of incidence at the boundary of two media is zero** i.e. when the ray of light passes through normal. b) When the **refractive index of both medium is same.**
Total internal reflection	a) The light must travel from a **denser to a rarer** medium. b) The angle of incidence must be **greater than the critical angle** for a pair of medium.
Hearing an echo	a) The minimum distance between the source of the sound and the reflector in air must be 17m. b) The size of the reflector must be larger than the wavelength of the sound waves. c) The intensity of the sound should be such that the reflected sound reaching the ear is sufficiently loud to be audible
Resonance	The frequency of the applied force must be equal to the natural frequency of the vibrating body.
Fuse wire (Alloy of lead and tin)	a) Low melting point b) High resistance so as to produce sufficient heat to melt it.
Calorimeter (made up of copper)	a) Good conductor of heat b) Low specific heat capacity
Substance to be radioactive	The number of neutron must be much larger than the number of protons.